Fundamental Theorem of Calculus

F18CE Week 8

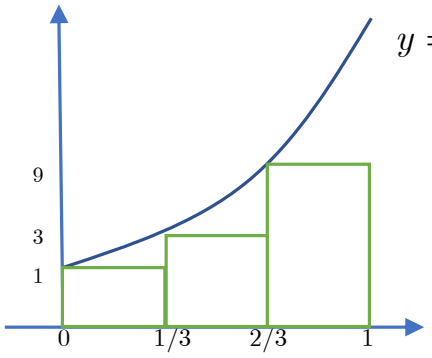
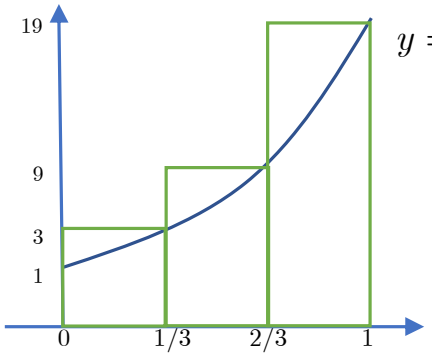
Recap

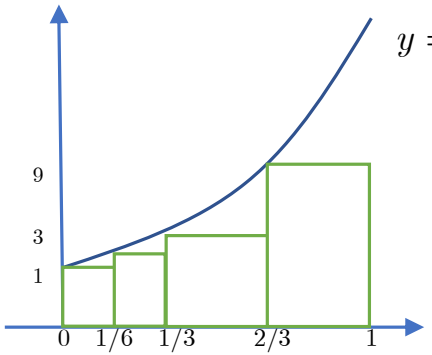
- Partition of $[a,b]$: $P = \{a = x_0, x_1, ..., x_k = b\}$, where $a = x_0 < x_1 < ... < x_k = b$.
- Length/gap of each subinterval: $\delta_r := x_{r+1} - x_r \quad \forall r = 0, 1, \ldots, k-1$.
- Max width of the subinterval: $\delta_{\max}(P) := \max\{\delta_0, ..., \delta_{k-1}\}$.
- Infimum of f in a subinterval: $I_r(f) = \inf\{f(x) : x \in [x_r, x_{r+1}]\}$ (view as $f_{\min}$ in $[x_r, x_{r+1}]$).
- Supremum of f in a subinterval: $S_r(f) = \sup\{f(x) : x \in [x_r, x_{r+1}]\}$ (view as $f_{\max}$ in $[x_r, x_{r+1}]$).

- Lower Riemann sum: $L(P,f)=\sum_{r=0}^{k-1} I_r(f)\delta_r$ (area under the graph formed by the lower rectangles).
- Upper Riemann sum: $U(P,f)=\sum_{r=0}^{k-1} S_r(f)\delta_r$ (area under the graph formed by the upper rectangles).
- Lemma For any partition P of $[a,b]$, $L(T_0,f)\le L(P,f)\le U(P,f)\le U(T_0,f)$

  where T_0 is the trivial partition of $[a,b]$, i.e., $T_0=\{a,b\}$.

- Refinement: A partition R is said to be a refinement of a partition P if $P \subset R$, that is, all the points in P are also in R.
- Lemma. Let P and R be partitions of $[a,b]$ such that R is a refinement of P. Then $L(P,f) \leq L(R,f)$ and $U(R,f) \leq U(P,f)$.
- Theorem. Let P and Q be any partitions of $[a,b]$. Then $L(P,f) \leq U(Q,f)$.

	$P = \{0, \frac{1}{3}, \frac{2}{3}, 1\}$ $L(P,f) = 1(\frac{1}{3}) + 3(\frac{1}{3}) + 9(\frac{1}{3}) = \frac{13}{3}$ $= 4.333$
	$P = \{0, \frac{1}{3}, \frac{2}{3}, 1\}$ $U(P,f) = 3(\frac{1}{3}) + 9(\frac{1}{3}) + 19(\frac{1}{3}) = \frac{31}{3}$ $= 10.333$ $* L(P,f) < U(P,f)$



$y = 18x^2 + 1$

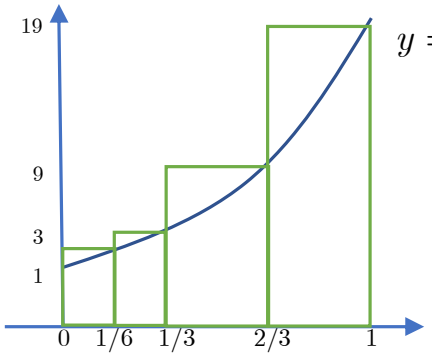
$R = \{0, \frac{1}{6}, \frac{1}{3}, \frac{2}{3}, 1\} \supset P$

R is a refinement of P

$L(R, f) = 1(\frac{1}{6}) + \frac{3}{2}(\frac{1}{6}) + 3(\frac{1}{3}) + 9(\frac{1}{3}) = \frac{53}{12}$

$= 4.41667$

$^*L(P, f) < L(R, f)$



$y = 18x^2 + 1$

$R = \{0, \frac{1}{6}, \frac{1}{3}, \frac{2}{3}, 1\} \supset P$

R is a refinement of P

$U(R, f) = \frac{3}{2}(\frac{1}{6}) + 3(\frac{1}{6}) + 9(\frac{1}{3}) + 19(\frac{1}{3}) = \frac{121}{12}$

$= 10.0833$

$^*U(R, f) < U(P, f)$

Definitions

Let $f : [a, b] \to \mathbb{R}$ be a bounded function.

Let P be an arbitrary partition of $[a, b]$.

Then we define the sets of lower/upper Riemann sum:

$$\mathcal{L}(f) := \{L(P, f) : P \text{ is a partition of } [a, b]\}$$

$$\mathcal{U}(f) := \{U(P, f) : P \text{ is a partition of } [a, b]\}.$$

Also, we define $\overline{L}(f) := \sup \mathcal{L}(f)$ and $\underline{U}(f) := \inf \mathcal{U}(f)$.

Example

Let $f(x) = x^2$. Let $P_n = \{0, \frac{1}{n}, ..., \frac{n-1}{n}, 1\}$ be a partition of $[0,1]$. Then

$$L(P_n, f) = \frac{1}{n}\left(\left(\frac{0}{n}\right)^2 + \left(\frac{1}{n}\right)^2 + ... + \left(\frac{n-1}{n}\right)^2\right) = \frac{1}{n^3}\frac{(n-1)n(2n-1)}{6} = \frac{(n-1)(2n-1)}{6n^2}$$

$$U(P_n, f) = \frac{1}{n}\left(\left(\frac{1}{n}\right)^2 + \left(\frac{2}{n}\right)^2 + ... + \left(\frac{n}{n}\right)^2\right) = \frac{1}{n^3}\frac{n(n+1)(2n+1)}{6} = \frac{(n+1)(2n+1)}{6n^2}$$

(Recall $\sum_{r=1}^{n} r^2 = \frac{n}{6}(n+1)(2n+1)$)

So, $\mathcal{L}(f) := \left\{ \frac{(n-1)(2n-1)}{6n^2} : n \in \mathbb{N} \right\}$ and $\mathcal{U}(f) := \left\{ \frac{(n+1)(2n+1)}{6n^2} : n \in \mathbb{N} \right\}$.

Hence, $\overline{L}(f) := \sup \left\{ \frac{(n-1)(2n-1)}{6n^2} : n \in \mathbb{N} \right\} = \lim_{n \to \infty} \frac{(n-1)(2n-1)}{6n^2} = \frac{1}{3}$, and

$$\underline{U}(f) := \inf \left\{ \frac{(n+1)(2n+1)}{6n^2} : n \in \mathbb{N} \right\} = \lim_{n \to \infty} \frac{(n+1)(2n+1)}{6n^2} = \frac{1}{3}.$$

(Theorem 7.5, Chap 2, F18CD)

Note that $\overline{L}(f) = \underline{U}(f)$ for this example. What is this means?

Definition 6.10 Integrability of function

A bounded function $f : [a,b] \to \mathbb{R}$ is said to be Riemann integrable on $[a,b]$ if there exist a partition of $[a,b]$ such that $\boxed{\overline{L}(f) = \underline{U}(f)}$, and we abbreviate the integration of f as

$$\int_a^b f\,dx \quad \text{or} \quad \int_a^b f\,,$$

where $\boxed{\int_a^b f\,dx = \overline{L}(f) = \underline{U}(f)}$.

Remark: f is not integrable if $\overline{L}(f) \neq \underline{U}(f)$ for any partition of $[a,b]$.

Remark: 1. This notation $\int_a^b f\,dx$ is useful if we need to emphasize which variable is being integrated, or we want to put in a specific function.

2. This notation $\int_a^b f$ is shorter and simpler if we are doing an abstract integration and have not defined the letter we are using for the variable of integration.

3. It is also convenient to define the integral of f when the lower limit is greater than, or even equal to, the upper limit, i.e., if $a < b$, then

$$\int_b^a f dx := -\int_a^b f dx\,.$$

4. The above constructions have been motivated by finding the area between the graph of f and the x-axis; and worked out the areas of rectangles by multiplying 'heights' by 'widths', where the 'heights' are given by function values.

5. So, if any function values are negative the corresponding 'areas' are negative. That is, the Riemann integral counts 'areas' below the x-axis as negative.

Recall Lemma 6.6

For any partition P of $[a,b]$, we have

$$L(T_0, f) \leq L(P, f) \leq U(P, f) \leq U(T_0, f).$$

Theorem 6.9

For any partitions P, Q,

$$L(P,f) \leq \overline{L}(f) \leq \underline{U}(f) \leq U(Q,f)\,.$$

Example 6.11 (Non integrable function)

Consider the function $f:[0,1]\to\mathbb{R}$ defined by

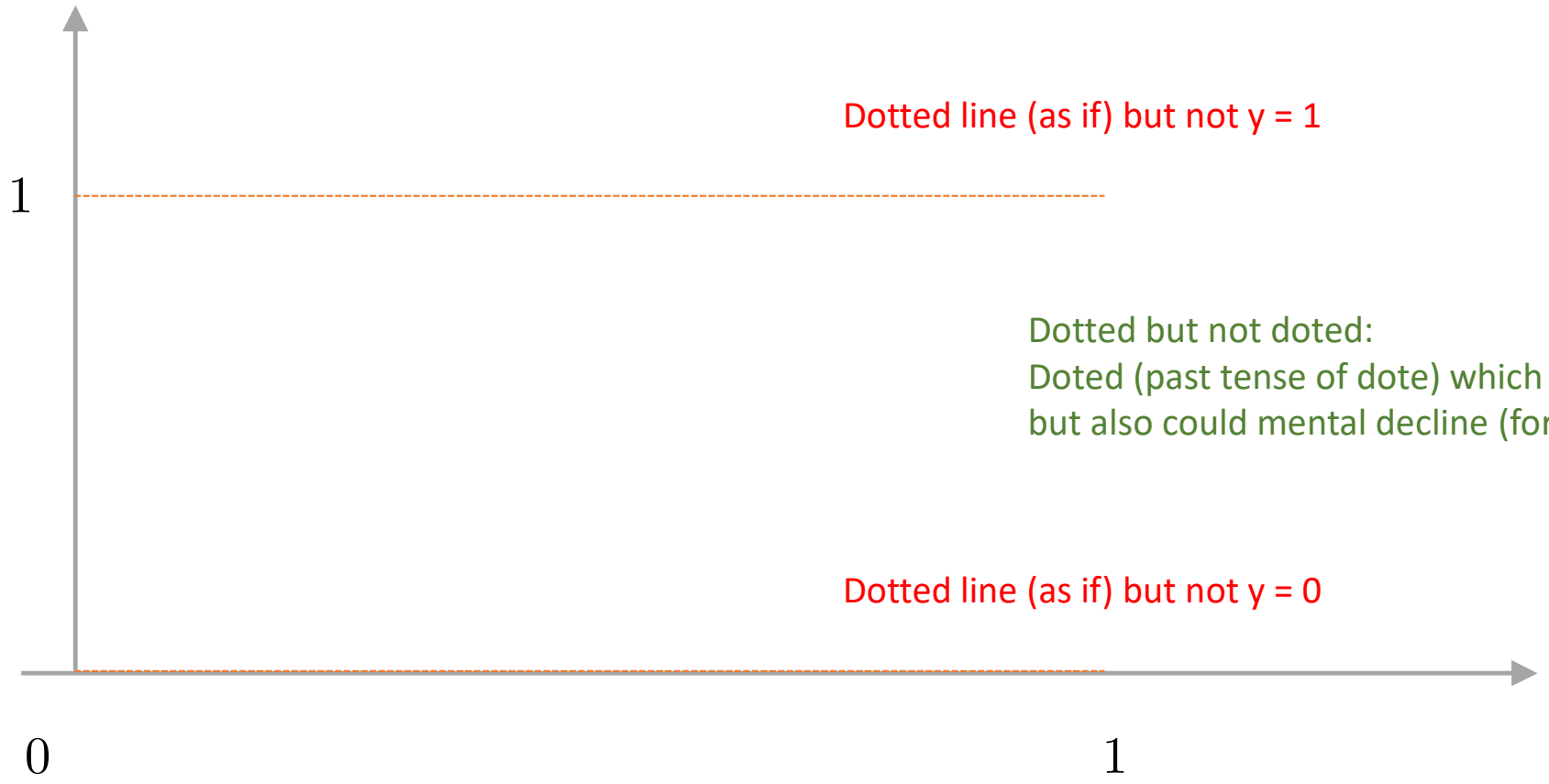
$$f(x)=\begin{cases}0, & x\notin\mathbb{Q}\\ 1, & x\in\mathbb{Q}\end{cases},\ x\in[0,1].$$

For any partition P, every interval $[x_r,x_{r+1}]$ contains both rational and irrational numbers, so we see that

$$I_r(f)=\inf\{f(x):x\in[x_r,x_{r+1}]\}=0\,,\ S_r(f)=\sup\{f(x):x\in[x_r,x_{r+1}]\}=1$$

If P is some partition of $[0,\ 1]$, then the density of the rationals in $\mathbb{R}$ implies that every subinterval of P will contain a point where $f(x) = 1$. It follows that $U(P, f) = 1$. On the other hand, $L(P, f) = 0$ because the irrationals are also dense in $\mathbb{R}$. Because this is the case for every partition P, we see that the upper Riemann sum is 1 but the lower Riemann sum is 0. The two are not equal, so we conclude that Dirichlet's function is not integrable.

The graph look likes this



Theorem 6.12

Let $f : [a,b] \to \mathbb{R}$ be a bounded function.

The following statements are equivalent:

Let P be an arbitrary partition of $[a,b]$.

(a) f is integrable on $[a,b]$.

(b) There exists a partition P_n of $[a,b]$ such that $\lim_{n\to\infty} U(P_n, f) = \lim_{n\to\infty} L(P_n, f)$.

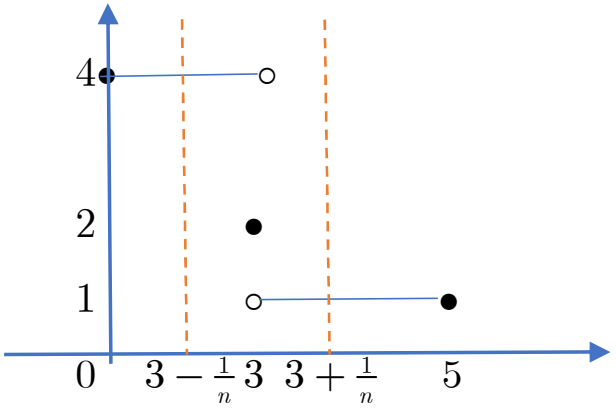
(c) There exists a partition P_n of $[a,b]$ such that $\lim_{n\to\infty} \Big(U(P_n, f) - L(P_n, f)\Big) = 0$.

Remark: $\int_a^b f\,dx = \lim_{n\to\infty} U(P_n, f) = \lim_{n\to\infty} L(P_n, f)$ if f is integrable on $[a,b]$.

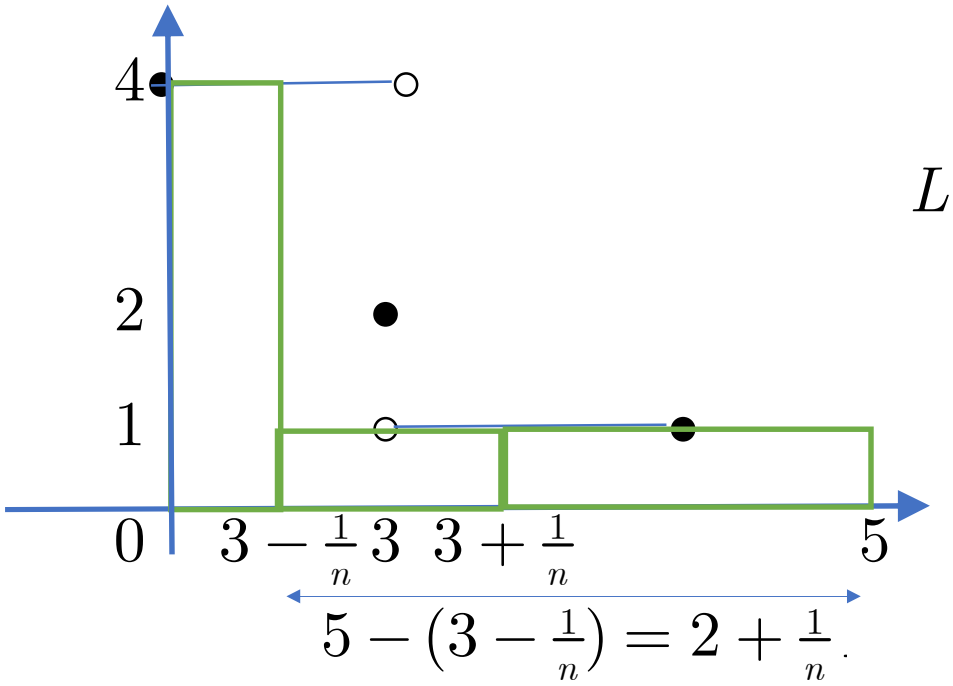
Example 6.13

Let $f:[0,5]\to\{1,2,4\}$ defined by

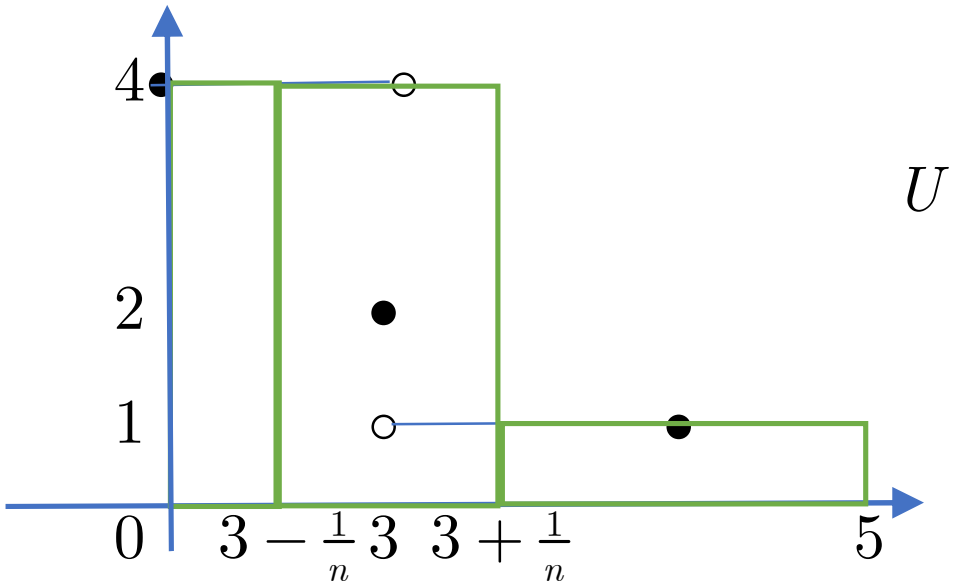
$$f(x)=\begin{cases}4, & 0\le x<3\\ 2, & x=3\\ 1, & 3<x\le 5\end{cases}.$$



Let $P_n=\{0,3-\frac{1}{n},3+\frac{1}{n},5\}$ be the partition of f for $n=1,2,3,\ldots$. Find $U(P_n,f)$ and $L(P_n,f)$ and show that f is integrable on $[0,5]$. Hence, find $\int_0^5 f\,dx$.



$$L(P_n, f) = 4(3 - \tfrac{1}{n}) + 1(2 + \tfrac{1}{n}) = 14 - \tfrac{3}{n}$$



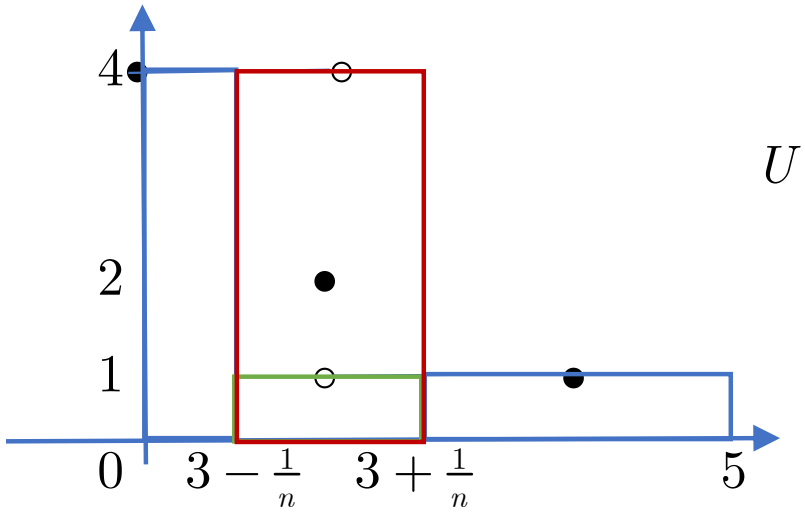
$$U(P_n, f) = 4(3 + \tfrac{1}{n}) + 1(2 - \tfrac{1}{n}) = 14 + \tfrac{3}{n}$$

$$L(P_n, f) = 4(3 - \tfrac{1}{n}) + 1(2 + \tfrac{1}{n}) = 14 - \tfrac{3}{n} \Rightarrow \lim_{n\to\infty} L(P_n, f) = 14$$

$$U(P_n, f) = 4(3 + \tfrac{1}{n}) + 1(2 - \tfrac{1}{n}) = 14 + \tfrac{3}{n} \Rightarrow \lim_{n\to\infty} U(P_n, f) = 14$$

Hence, $\lim_{n\to\infty} L(P_n, f) = 14 = \lim_{n\to\infty} U(P_n, f)$. So f is integrable on $[0, 5]$ and $\int_0^5 f(x)\,dx = 14$.

Alternatively, we can find $U - L$ directly from the graph to show that f is integrable on $[0, 5]$.



$$U(P_n, f) - L(P_n, f) = 3(\tfrac{2}{n}) = \tfrac{6}{n} \Rightarrow \lim_{n\to\infty}\Big(U - L\Big) = 0$$

Theorem 6.14: Properties of integral

Let $f, g : [a,b] \to \mathbb{R}$ be integrable functions on $[a,b]$. Let c be a constant. Then

(a) $f+g$ is integrable on $[a,b]$, with $\int_a^b (f+g) = \int_a^b f + \int_a^b g$.

(b) cf is integrable on $[a,b]$, with $\int_a^b (cf) = c\int_a^b f$.

(c) fg is integrable on $[a,b]$, with $\left(\int_a^b (fg)\right)^2 \leq \left(\int_a^b f^2\right)\left(\int_a^b g^2\right)$ (Cauchy's inequality)

(d) $\left|f\right|$ is integrable on $[a,b]$, with $\left|\int_a^b f\right| \leq \int_a^b \left|f\right|$ (analogous to triangle inequality)

(e) $f(x) \leq g(x)\ \forall x \in [a,b]$ implies that $\int_a^b f(x)dx \leq \int_a^b g(x)dx$.

(f) $f(x) = c\ \forall x \in [a,b]$ implies that $\int_a^b f(x)dx = c(b-a)$.

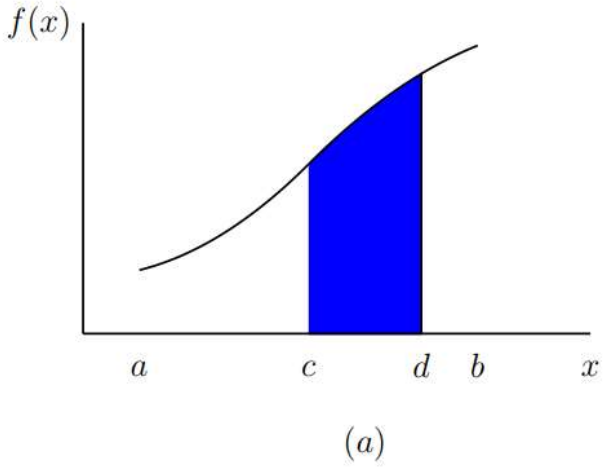
Theorem 6.14, Theorem 6.15

Let $f : [a,b] \to \mathbb{R}$ be integrable functions on $[a,b]$.

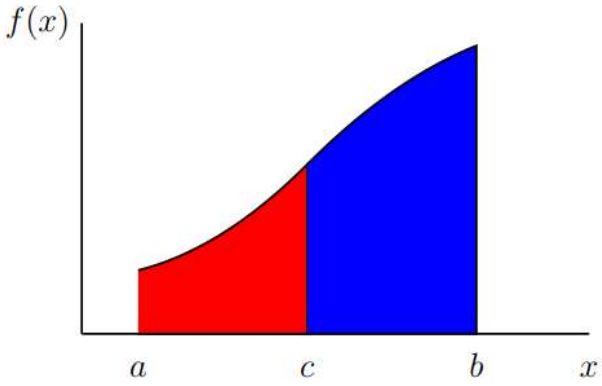
(a) If $[c,d] \subset [a,b]$, then f is integrable on $[c,d]$.

(b) If $a \le c \le b$, then $\int_a^b f = \int_a^c f + \int_c^b f$.

(c) $\int_c^c f = 0 \quad \forall c \in [a,b]$.



(a)



(b)

Theorem 6.16 & Theorem 6.17

Let $f : [a,b] \to \mathbb{R}$ be a bounded function.

- If f is monotone on $[a,b]$, then f is integrable on $[a,b]$.
- If f is continuous on $[a,b]$, then f is integrable on $[a,b]$.
- If f has a finite number of jump discontinuities on $[a,b]$, then f is integrable on $[a,b]$.

Remark: The converse of the each of the above statements is not true in general.

Corollary 6.18

Suppose f is integrable on $\left[a,b\right]$. Let $\left(P_n\right)=\left(P_1,P_2,...P_k,...\right)$ be a sequence of partitions such that $\delta_{\max}\left(P_n\right)\to 0$. Denote these partitions by

$$P_n=\left\{x_{n,r}:r=0,1,...,k_{n-1},n\in\mathbb{N}\right\}\subset\left[a,b\right],$$

i.e., among all these partitions (note that $a=x_{n,0}$ and $b=x_{n,k_{n-1}}\ \forall n\in\mathbb{N}$):

$$P_1=\left\{x_{1,0},x_{1,1},...,x_{1,k_0}\right\},P_2=\left\{x_{2,0},x_{2,1},...,x_{2,k_1}\right\},...,P_n=\left\{x_{n,0},x_{n,1},...,x_{n,k_{n-1}}\right\},...,$$

the max gap between each subinterval is close to 0.

Suppose $c_{n,r} \in \left[x_{n,r}, x_{n,r+1}\right]$ and $\delta_{n,r} = x_{n,r+1} - x_{n,r}$. Then

$$\int_a^b f = \lim_{n\to\infty} \sum_{r=0}^{k-1} f(c_{n,r})\delta_{n,r} .$$

Remark

Remark: In other words, if f is integrable on $[a,b]$. Then we are free to make any choice on the 'height' of each subinterval, i.e., any arbitrary $x^* \in [x_k, x_{k+1}]$ can be chosen to obtain $f(x^*)$ (for all k), no necessarily the sup/max or the inf/min of f in each subinterval.

Theorem 7.1 Fundamental Theorem of Calculus

(i) If $f : \left[a,b\right] \to \mathbb{R}$ is integrable on $\left[a,b\right]$, and $F : \left[a,b\right] \to \mathbb{R}$ satisfies $F'(x) = f(x)$ for all $x \in \left[a,b\right]$, then $\boxed{\int_a^b f(x)\ dx = F(b) - F(a)}$.

(ii) Let $g : \left[a,b\right] \to \mathbb{R}$ be integrable on $\left[a,b\right]$ and define $G(x) = \int_a^x g(t)\ dt$ for all $x \in \left[a,b\right]$. Then, G is continuous on $\left[a,b\right]$. If g is continuous at some point $c \in \left[a,b\right]$, then G is differentiable at c and $\boxed{G'(c) = g(c)}$.

We provide the proof for (i)

Let P be a partition of $\left[a,b\right]$ and apply the Mean Value Theorem to F on a typical subinterval $\left[x_{k-1},x_k\right]$ of P. This yields a point $t_k \in \left(x_{k-1},x_k\right)$ where

$$F(x_k) - F(x_{k-1}) = (x_k - x_{k-1})F'(t_k) = (x_k - x_{k-1})f(t_k)\,.$$

Now, consider the upper and lower sums $U(P,f)$ and $L(P,f)$. Since $m_k \leq f(t_k) \leq M_k$ (where m_k is the infimum on $\left[x_{k-1}, x_k\right]$ and M_k is the supremum), it follows that

$$L(P,f) \leq \sum_{k=1}^{n} \Big[F(x_k) - F(x_{k-1})\Big] \leq U(P,f).$$

But notice that the sum in the middle is telescopic, i.e.,

$$\sum_{k=1}^{n}\Big[F(x_{k}) - F(x_{k-1})\Big] = F(b) - F(a)\,,$$

which is independent of the partition P. Thus, we have

$$L(P,f) \leq F(b) - F(a) \leq U(P,f)\,.$$

Because $\overline{L}(f) = \underline{U}(f) = \displaystyle\int_a^b f$, we conclude that $\displaystyle\int_a^b f = F(b) - F(a)\,.$

Example (a)

Let $F(x) = \int_x^{\sin x} \frac{1}{2+t^2}\, dt$. State why F is differentiable and compute $F'(x)$.

Proof. By FTOC, if $g(t)$ is a continuous function on a closed interval $[a,b]$,

then $G(x) := \int_a^x g(t)\, dt$ is differentiable for all $x \in \Big[a,b\Big]$ and $G'(x) = g(x)$.

Let $g(t) = \frac{1}{2+t^2}$. Then g is continuous everywhere since $2 + t^2 \neq 0 \ \forall t \in \mathbb{R}$.

Moreover, $F(x) = \int_x^{\sin x} \frac{1}{2+t^2}\, dt = \int_a^{\sin x} \frac{1}{2+t^2}\, dt - \int_a^{x} \frac{1}{2+t^2}\, dt = G(\sin x) - G(x)$

$$\Rightarrow F'(x) = (\cos x)\ G'(\sin x) - G'(x) = \frac{\cos x}{2+\sin^2 x} - \frac{1}{2+x^2}\,.$$

Also, since G is differentiable, F is differentiable.

Example (b)

Let $g:\left[0,1\right]\to\mathbb{R}$ be a function defined by $g(x)=\frac{1}{x+1}$.

Let $n\in\mathbb{N}$ and $P_n=\{0,\frac{1}{n},\frac{2}{n},...,\frac{n-1}{n},1\}$ be a partition.

(a) Show that g is Riemann integrable on $\left[0,1\right]$.

(b) Use the Lower Riemann sum in (a) to prove that

$$\lim_{n\to\infty}\left(\frac{1}{n+1}+\frac{1}{n+2}+...+\frac{1}{2n-1}+\frac{1}{2n}\right)=\ln 2\,.$$

Solution. (a) $U(P_n, g) = \frac{1}{n}\left(\frac{1}{0+1} + \frac{1}{\frac{1}{n}+1} + \ldots + \frac{1}{\frac{n-1}{n}+1}\right)$,

$$L(P_n, g) = \frac{1}{n}\left(\frac{1}{\frac{1}{n}+1} + \frac{1}{\frac{2}{n}+1} \ldots + \frac{1}{\frac{n-1}{n}+1} + \frac{1}{1+1}\right).$$

$$U(P_n, g) - L(P_n, g) = \frac{1}{n}\left(1 - \frac{1}{2}\right) = \frac{1}{2n} \Rightarrow \lim_{n\to\infty}\Big(U(P_n, g) - L(P_n, g)\Big) = 0$$

and so g is integrable on $[0, 1]$.

(b) $L(P_n, g) = \frac{1}{n}\left(\frac{1}{\frac{1}{n}+1} + \frac{1}{\frac{2}{n}+1} \ldots + \frac{1}{\frac{n-1}{n}+1} + \frac{1}{1+1}\right) = \sum_{r=1}^{n} \frac{1}{n+r}$.

Since g is integrable on $[0, 1]$,

it follows that $\lim_{n\to\infty}\left(\sum_{r=1}^{n} \frac{1}{n+r}\right) = \int_0^1 \frac{1}{x+1}\, dx = \Big[\ln(x+1)\Big]_0^1 = \ln 2$.